

ALM25

Action LAB Mexico

- Aalto Team Edition -

Working with Solar Technologies and Assessing their Impact

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Sustainable Global Technologies Studio Course

WAT-E2070



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And finally, to the community of El 20: thank you for welcoming us so openly and warmly, for sharing your stories and for making us feel like part of your family.

List of Abbreviations

SGT - Sustainable Global Technologies
El20 - Ejido 20 de Noviembre
ALM - Action Lab Mexico
TFK - Team Finland Knowledge program
DYA - Design Your Action NGO
UAM - Universidad Autónoma Metropolitana

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Abstract

The Action LAB Mexico (ALM) 2025 continued Aalto University's long-standing collaboration under the Sustainable Global Technologies (SGT) Studio course. Founded by Claudia Garduño in 2012, ALM has been supported by the Team Finland Knowledge (TFK) programme, with partners working together for over 13 years. ALM 2025 is collaborating with Universidad Modelo, Universidad Autónoma Metropolitana (UAM), Design Your Action NGO (DYA), and the Mayan community Ejido 20 de Noviembre (EI20). Located in Calakmul, Campeche, on the Yucatán Peninsula, EI20 is the only Mayan community in the region. The 2025 team traveled to Mexico from March 14 to March 31, 2025, to conduct fieldwork.

Since the ALM project's first visit to EI20 in 2012, the community members have been the primary rights-holders. This year the ALM 2025 project's collective objective including all the teams is 'Co-Design for sustainable Transition of EI20's Nutrition-Energy and Socio-Technical System'. The broad project topics this year were solar ovens and solar cooking, nutrition and health of the community, social perceptions, water purification and energy. Key topics that were included in the projects were, perception, social development, innovation, sustainability, adaptability, water security, electrical analysis, and renewable energies. The ALM 2025 had multiple teams that focused on various aspects of promoting sustainable practices in the community.

This year the ALM Aalto Team has integrated three different ALM 2025 projects to work on to promote sustainable practices and enhance community well-being. The first project centers on solar cooking and how to integrate traditional Mayan cooking practices and recipes to be used with the solar oven. The second project involves collaboration between one Aalto team member and students from Universidad Autónoma Metropolitana (UAM) on a solar water distillation project. This project uses solar energy to purify water by reducing its mineral content, aiming to provide sustainable solutions to improve water security and provide the community with clean, safe drinking water. The third project focuses on social perceptions, and consists of perception interviews that aim to connect the solar cooking and water purification themes together. The interviews aimed to gather local residents' insights on earlier student projects involving solar ovens and water purification, helping future teams better align their work with the community's perspectives and needs.

Building on the ALM team 2021's strategy to create a Mayan women's cookbook, the solar oven cooking tests data were compiled into a cookbook-style zine designed for the community. In the zine, there are also some benefits from solar cooking that include insights received from the perception interviews. In addition, the results from the solar water distilling are reported and documented for future work and to be shared with the community. The results also provide up-to-date information of the condition of the tested water resources in the community.

Intro

1 Strcuture

This report is structured to reflect the chronological flow of our project: before, during, and after the field trip to Mexico. We chose this format to clearly illustrate how the project evolved over time and to make it easier for the reader to follow our process, decisions, and learning journey.

The first main section, *Before the Field Trip*, outlines the project's background, including an overview of the ALM initiative, information about the El 20 community, and previous work conducted there. It also covers our early research and planning activities, focusing on collaboration with local and international partners and the development of our topic and research focus. The section concludes with a description of our project planning, including stakeholder mapping, communication strategies, monitoring plans, a timetable, and other logistical preparations.

During the Field Trip covers our field activities, including the methodologies we applied, such as interviews and participant observation as well as our collaboration and communication with the community, and the testing and workshops conducted. This section also highlights our team dynamics, daily reflections, and the cultural experiences that shaped our journey.

After the Field Trip presents our data analysis, results, and reflections. We share findings from our project activities, describe our creative output (the zine), and reflect on the challenges, learnings, and overall impact. The report concludes with a discussion and outlines the next steps for future development.

2 Team



Zhengzhen

Zhengzhen is a Master student of EIT Urban Mobility at Aalto University and KTH Royal Institute of Technology. She is passionate about identifying issues of inequality, advocating for transport accessibility, and interested in different cultures. With work experience in urban planning and consulting, she aims to promote equitable and sustainable development.



Petra

Petra is a Master student in the Water and Environmental Engineering Master's Program at Aalto University. Her interests include environmental well-being and climate change mitigation as well as exploring ways to enhance sustainability in different contexts. In the project, she is motivated to find solutions that promote sustainability while respecting cultural heritage.



Johanna

Johanna is a Master student of Collaborative and Industrial, focusing on Social and Service Design at Aalto University. She is committed to sustainable development and design that is rooted in understanding cultural, social, and environmental contexts to create solutions that drive lasting change.



Ran

Ran is a Master student of EIT Urban Mobility at Aalto University and KTH Royal Institute of Technology. Studying sustainable transition, he wants to learn more than urban mobility. He always has passion for new adventures and new experiences, so he tried working in construction, consulting and civil aviation. He loves cultural heritage, and he hopes to dig new attractions while maintaining sustainability.



Best (Mentor)

Best is a doctoral researcher in the Water and Development Research Group. His expertise is in analysing global food system using global data. He also has experiences in development projects with local communities.



Pragati (Mentor)

Pragati is a researcher and mentor at the Sustainable Global Technologies group (SGT). A designer by profession, Pragati holds experience in empirical research and of designing and conducting participatory and interview based research. Her research interests lie at the intersection of sustainability, emotions and design through the lens of decolonial framework.

Before the Field Trip

3 Learning about ALM, EI20 and previous projects

Our journey began in early January with the start of the SGT course, where we got to know each other within our team, met our mentors, and familiarized ourselves with the foundation and context of our future project. Through background research and by reading project reports from previous student teams, we began to understand the initiative known as Action Lab Mexico (ALM), its long-standing collaboration between Aalto University and local partners in Mexico, and our potential role within this intercultural initiative.

ALM 2025

Action Lab Mexico (ALM) was founded in 2012 by Claudia Garduño as part of her doctoral dissertation Design for Freedom (Garduño, 2017), in collaboration with the Mayan community of El 20 in the Calakmul region of Campeche. Since then, ALM has become an annual component of the Sustainable Global Technologies (SGT) Studio Course at Aalto University. Each year, it brings together interdisciplinary student teams from Finnish universities and Mexican universities to co-develop context-sensitive solutions with the community of El 20, guided by a human-centered and collaborative approach. ALM is supported by the Team Finland Knowledge (TFK) program of the Finnish National Agency for Education (ALM co-mentora, 2023–2025).

EI 20

El 20 de Noviembre is a small Mayan community in the municipality of Calakmul, in the state of Campeche on the Yucatán Peninsula, Mexico (Figure 1). As of 2020, the community had a population of approximately 419 residents (Mexico PueblosAmerica, 2024). It is believed to have been founded in 1971 by families who migrated from the Mayan town of Dzitbalché (Garduño García, 2017), and it remains the only Mayan community in the Calakmul region.

El 20 is located near major archaeological sites, including the ancient Mayan city of Calakmul, the largest in Latin America, and the recently discovered site of Valeriana, which is expected to become the second-largest (Rannard, 2024).

Despite its rich cultural heritage and natural surroundings, the region faces significant environmental challenges. Extended droughts have severely depleted local water sources and affected both agriculture and daily life (Buchanan, 2020). Wildfires are also a recurrent threat, such as the 2023 fire in the Balam Kiin ecological reserve that destroyed 400 hectares of grassland (Magaña, 2023).

Daily life in El 20 reflects both cultural resilience and socio-economic constraints. The community engages in activities such as artisanal handicrafts, Melipona beekeeping, agriculture, and small-scale tourism. Traditional crafts, including woodworking, jewelry, textiles, and hammock weaving, play a vital role in both cultural preservation and income generation (ALM Report, 2023). Honey production from Melipona bees, a practice with deep roots in Mayan culture, is valued for both its economic and medicinal significance. While agriculture remains mostly for subsistence, water-related challenges such as hard water and reliance on rainwater make farming increasingly difficult (ALM Report, 2023). Access to healthcare is also limited; nearby clinics are under-resourced, and the national healthcare system does not cover treatment at the closest hospital in Chetumal. Some residents suffer from kidney stones, potentially linked to high mineral content or bacterial contamination in well water (ALM Report, 2023).



Figure 1: Map of Campeche

Previous Projects

Teams from previous years have explored various aspects of Mayan heritage, including architecture and handicrafts, while also addressing key development issues such as healthcare, economic autonomy, sustainable tourism, and access to clean water.

While the topics have been diverse, water-related work has been a recurring theme. The 2013 team explored water purification techniques such as boiling, filtration, and rainwater harvesting (ALM'13 Team, 2013). Later teams expanded these efforts, conducting water quality tests for *E. coli*, ammonia, and nitrate, and experimenting with innovations including filters (ALM'18 Team, 2018), fog collectors (ALM'22 Team, 2022), solar distillers (ALM'22 & ALM'23 Teams, 2022-2023), and biofilters (ALM'23 Team, 2023). In 2021, a consolidated Water Strategy was compiled to guide and evaluate ongoing water-related work (ALM'21 Team, 2021).

Beyond water, past teams have contributed to economic and cultural development, especially by supporting traditional crafts and community-based tourism. The brand “Artesanía para el Bienestar” (ALM Report, 2016) was created to promote local handcrafted goods, including stone carvings, woven items, and natural seed jewelry. These products are sold locally and via social media, though reach and engagement remain relatively limited (ALM Report, 2024). Tourism-related infrastructure includes an eco-hostel, designed in 2013 and completed in 2015 (ALM Report, 2015), which provides sustainable lodging and supports tourism initiatives such as bird-watching (ALM'20 Team, 2020). The 2021 team also focused on amplifying women's voices through storytelling and the documentation of traditional culinary knowledge (ALM'21 Team, 2021). Most recently, in 2024, student teams at Aalto University designed and tested solar ovens, which were later introduced to the community with promising results (ALM'24 Team, 2024a).

All this prior work provided a solid foundation for us to build upon. It helped us identify persistent challenges and understand community priorities. Reading the stories of past projects not only gave us inspiration, but also instilled a sense of responsibility. We became aware of the depth and continuity of this ongoing process and began thinking critically about how we could contribute meaningfully to its next chapter.

4 Developing our project topic(s)

Our project development process began with a clear intention: define one concrete topic as the Aalto team. Spoiler: we didn't end up with one, but with three. After reviewing community needs, past project gaps, and our own skills and interests, we began shaping two main project ideas: Waste Management and Tourism with a focus on Traditional Cooking. Later, a third theme, Solar Cooking, emerged as an intersection of both and as a continuation of ALM 2024's solar oven efforts. But that still wasn't the end. At a later stage, Solar Distilling and the question of community perception of past projects also entered the picture. But let's start from the beginning.

Our first Direction

Our topic-finding process was iterative. We brainstormed, mapped ideas, and continuously exchanged thoughts with our mentors.

Waste Management was identified as a pressing issue as early as 2012 during the first ALM project, which proposed community-level strategies to reduce, separate, and reuse waste (Report, 2012). Yet over the past decade, waste has remained a visible challenge. The 2023 team observed that waste was still being disposed of on roadsides and burned, which raises serious environmental and health concerns (Report, 2023). We began by asking questions such as: What types of waste are generated in El 20, and in what quantities? What existing knowledge or infrastructure could support reuse, recycling, or composting? How aware are residents of the risks associated with waste burning? And most importantly: How can any new system be simple, affordable, and community-driven? With these questions in mind, we framed our goal as: "Improving the waste management system within the community for health and environmental wellbeing."

Tourism & Traditional Cooking was our second topic, building on El 20's existing tourism potential. As the only Mayan community in the region, it offers authentic cultural experiences from local crafts to nature-based tours. While several past ALM teams supported tourism infrastructure and branding, such as the Artesanía para el Bienestar label or the eco-hostel, none had specifically focused on traditional food as a tourism experience. We saw an opportunity to explore culinary tourism in a way that could both create income and strengthen cultural pride. Our guiding questions were: What are the traditional cooking practices of El 20, and how can they be shared with visitors? How can we co-create a tourist experience that is respectful, interactive, and rooted in Mayan culture? How can such initiatives be marketed, and how can tourism be developed without harming the environment? We defined our intended impact as: "Promoting economic growth through sustainable tourism centered on preserving and sharing traditional cooking culture."

Evolving our Direction through Collaboration with Modelo and UAM

In February, our collaboration deepened as we began weekly joint meetings with students from Universidad Modelo and UAM, along with their mentors Valeria Inés Canto Farjat, Miguel Santiago Pérez Centeno, and professor Silvia Barrera Suárez, and ALM founder Claudia Garduño García. These meetings became a crucial space to share ideas, align expectations, and co-develop project directions across teams. Through these conversations, we gained valuable insights into the local context. For example, while we initially saw Waste Management as a strong project direction, we were informed that the community had already begun composting practices. Similarly, we learned that El 20 already offers basic cooking workshops for tourists. This made us question our project ideas.

At the same time, a new direction started to emerge: Solar Cooking. One student team from Modelo had been asked to continue the solar oven project started in 2024, and since we had already explored traditional cooking, the idea of combining it with solar technology felt like an interesting evolution of our project.

Shifting our focus to solar ovens

Solar ovens offer benefits across health, economy, and the environment. They provide smoke-free cooking, reducing respiratory irritation and health risks. They are cost-effective, using free solar energy while saving on firewood, charcoal, gas, or electricity. Environmentally, they help preserve forests and reduce carbon emissions. Additionally, solar ovens are safe, easy to use, and require minimal supervision, freeing up time for other activities (Solar Cookers International, 2004).

Building on the 2024 project, where the Aalto team constructed a solar oven and brought it to the community, this year's project aims to collaborate with the Mexican team to expand solar oven usage and make solar cooking more accessible. While the Modelo and UAM teams focused on constructing additional ovens using local materials and organizing hands-on workshops, our team wanted to collaborate closely with local women to explore how traditional Mayan recipes can be adapted for use in solar ovens, ensuring the technology is practical and beneficial for daily life.

Through interviews, participatory activities, testing, and workshops, we planned to gather insights that would be compiled into a cookbook-style leaflet, zine, or brochure tailored for the community. Additionally, we considered the potential of selling this resource to tourists as a cultural souvenir, creating an opportunity for the community to generate extra income. We defined our project's impact as: "Integrating Solar Ovens into Traditional Mayan Cooking for Sustainable Practices and Community Well-being by increasing the confidence of the local community in using solar ovens for their traditional recipes." Our ultimate goal is for the local community to adopt solar ovens as a regular cooking method, leading to long-term health and environmental benefits. At the same time, we hope to promote economic autonomy by enabling the community to generate income from selling the cookbook-style resource to tourists.

Again, some changes

But this still wasn't the end. Only a couple of weeks before traveling to Mexico, another shift occurred. Through continued discussions with the Mexican teams, it became clear that there were multiple parallel project opportunities for collaboration.

From these conversations, a new path emerged: Petra, from the Aalto team, joined a project focused on water purification, working with students from UAM. Her final topics became solar distillation and perception interviews. The goal of the perception interviews was to form a bridge between the Aalto team's different topic areas by gathering local perspectives on both solar ovens and water-related student projects. Meanwhile, the rest of the Aalto team (Johanna, Ran, and ZZ) continued to focus on the solar cooking project.

5 Getting ready for the field trip

Preparing for the field trip to Mexico involved not only finalizing our project topic but also required thorough planning on multiple levels, from logistics to research preparation.

One of our first key deliverables for the course was our project plan, which served as an essential tool to structure our work and guide our collaboration across the months. It included a detailed timeline for both the course and the project (Figure 2), as well as preliminary versions of our fieldwork and budget plans. While those changed later during the process and are not shown here, the next section "During the fieldtrip" presents our actual day-by-day fieldwork plan reflecting what we really did on-site.

To support project readiness, we conducted a risk analysis in advance. This helped us identify both internal and external risks, assess their potential impact, and outline mitigation strategies. Additionally, we prepared a monitoring and evaluation plan to track our progress and ensure we remained aligned with our objectives. We also evaluated the sustainability of our project topics to guide our thinking and shape the topics and fieldwork plans, and the evaluation was updated in the later stages of the project as well. A communication plan was also created to ensure smooth coordination between all team members and stakeholders throughout the project.

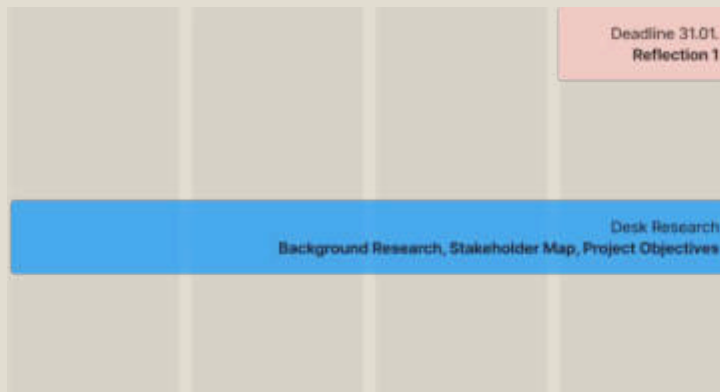
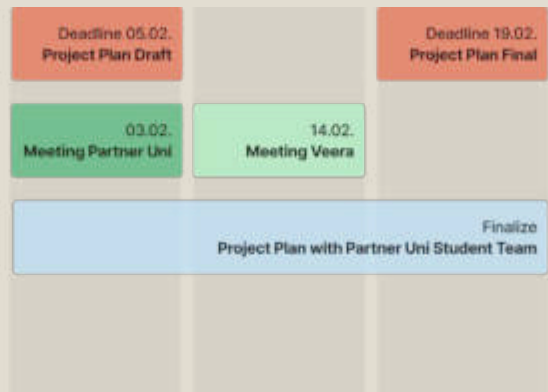
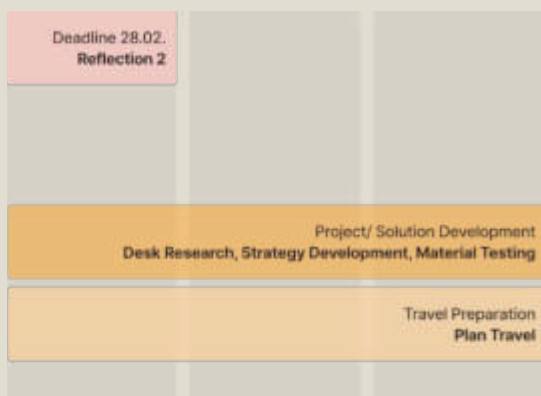
January**February****March****April****May**

Figure 2: Timetable

Stakeholder Mapping

One of the most important preparatory activities was our stakeholder analysis, which we have updated for this report (Figure 3). We categorized stakeholders into seven groups: community, academia, NGOs, public sector, funding bodies, tourists, and more-than-human entities. Each group holds varying levels of power and interest in the project, as visualized in the map.

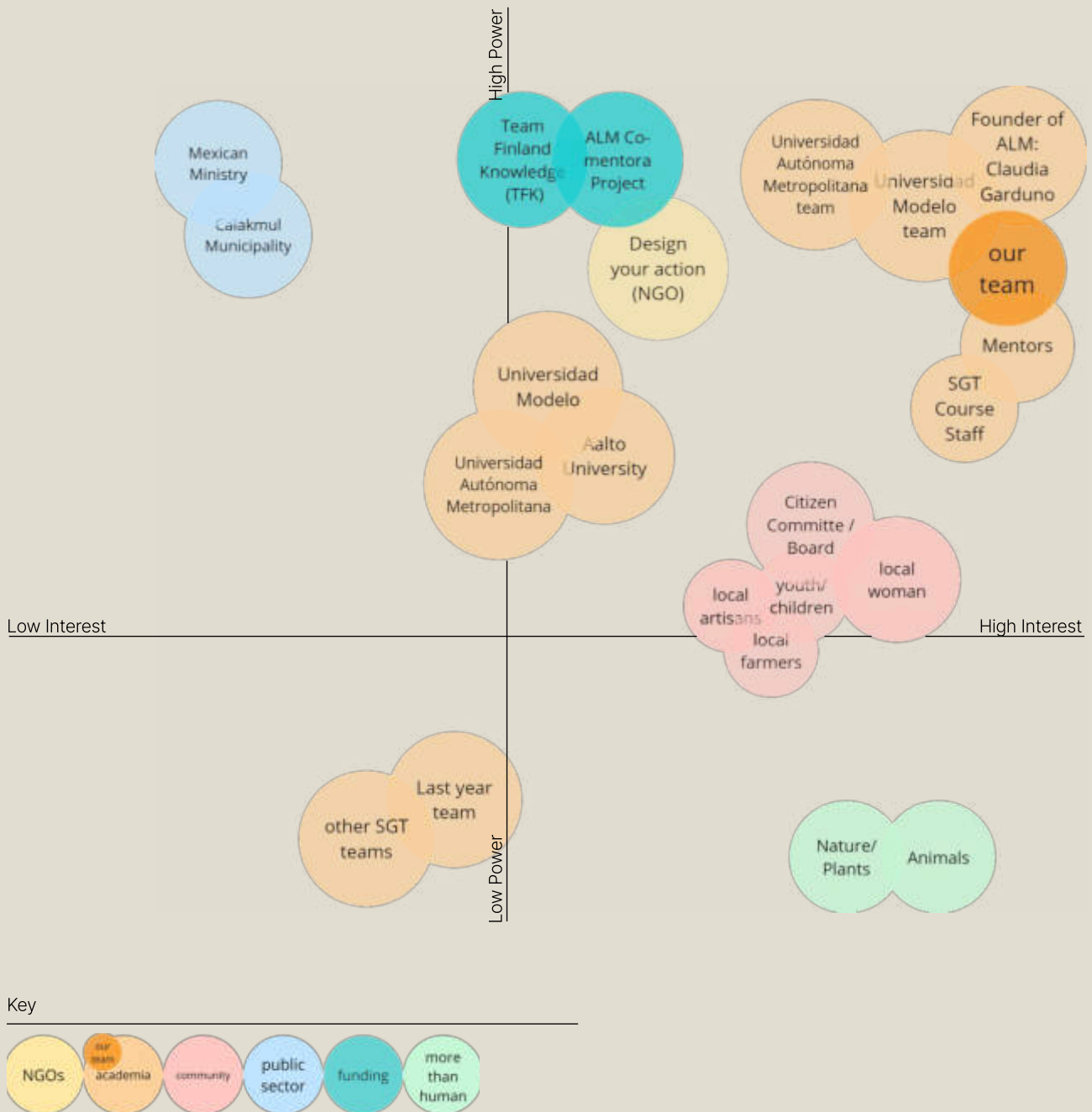


Figure 3: Stakeholder Map

The stakeholders in the ALM 2025 project are divided into six different groups, which are community, academia, NGO, public sector, funding and more than human. All the stakeholders have different powers and interests in the project as visualized in the map (Figure 2).

Community: The target community Ejido de 20 Noviembre is the main stakeholder of the project. Instead of as a whole, the different groups of people including local artisans, local farmers, youth/children, citizen committee/board, and local people working in relevant areas can better represent the community. They directly engage in interviews, workshops and pilot activities, providing local knowledge and insights into community needs. The project implementation will be carefully designed to align with their specific priorities. The success of the project may improve health, and culinary culture protection. They are positioned as high-interest, moderate-power because their influence relies on collaboration with NGOs and academia due to limited resources.

Academia: Aalto University students, mentors, Universidad Modelo teams, and ALM founder Claudia Garduño will play key roles in leading and collaboratively participating in research, planning, and implementation. Universidad Modelo's local team has a strong understanding of the local environment, helping to mitigate risks such as language barriers, which gives them greater influence than the Aalto team. Students, mentors, and staff will gain practical skills through their involvement, positioning them as high-interest, high-power stakeholders whose technical expertise will directly shape project outcomes. Meanwhile, institutions like Aalto University and Universidad Modelo have management rights over these individuals but are not directly engaged in the project's execution, classifying them as low-interest, high-power stakeholders.

NGOs: NGOs such as Amigos de Calakmul (ACAC), Design Your Action (DYA), Waterfinns, and others play a crucial role in community empowerment by contributing their expertise and experience. Their extensive networks facilitate the efficient implementation of pilot projects and workshops. As moderate-interest, moderate-power stakeholders, they can provide valuable resources, support policy initiatives, and strengthen the project's reach through their partnerships.

Public Sector: The Mexican Ministry and Calakmul Municipality influence the project indirectly such as regulations about stove specifications. Successful projects could contribute to regional sustainability strategies. However, since they will not take part in the ALM project itself, they are considered low-interest, high-power stakeholders due to their regulatory authority.

Funding: The ALM project is funded and supported by the Team Finland Knowledge (TFK) program of the Finnish National Agency for Education (ALM co-mentora 2023–2025), which enables fieldwork and prototype development. TFK's requirements guide and influence the project, ensuring it aligns with their broader mission. While TFK is medium-interest, as they are not focused on the specific topic of our project but rather on its overall success, their high-power comes from their control over funding, which is essential for the project's viability and progress.

Tourists: Tourists and travel agencies are important stakeholders in the project, as their spending supports the local community's income, and their behavior impacts the community environment. Positioned as moderate-interest, medium-power stakeholders, their influence is indirect but still significant, as their actions and choices can affect the success of the project's sustainability goals and the long-term well-being of the community.

More-Than-Human: Nature plants, animals and ecosystems are passive stakeholders. Environmental risks threaten project success, and sustainable practices protect ecosystems. The project need to adopt sustainable solutions, minimize environmental harm, and promote coexistence with the natural world. Although positioned as low-interest, low-power, their needs require mandatory integration into the whole process of the project.

Rights-holders

Additionally, we assessed the rights-holders within our project. Rights-holders are the individuals and organizations who benefit from the project. The right holders of this project are divided into two categories: direct rights-holders and final rights-holders. Direct rights-holders are usually individuals and organizations who are the direct recipients of cooperation. Final rights-holders are usually individuals and organizations who benefit from the project in the long term.

Direct rights-holders:

The team members of ALM 2025: The ALM 2025 team consists of four students and two mentors. Beyond earning reports, credits, and grades, the team gains unique and valuable experiences through hands-on project work and immersive travel exploring Mexico's Mayan culture. This project strengthens their critical thinking, collaboration, and problem-solving skills in a dynamic, multidisciplinary, and intercultural environment.

Partner Team Members from Universidad Modelo and Universidad Autónoma Metropolitana: The partner team consists of 19 students, 2 mentors and 2 professors. The ALM project founder, Dr. Claudia Garduño García, and 5 students are from Universidad Autónoma Metropolitana. The remaining 14 students, 2 mentors and professor Silvia Barrera Suárez are from Universidad Modelo. The partner team from Universidad Modelo and Universidad Autónoma Metropolitana work together with the Aalto team. We engaged in learning, meeting and workshops before the field trip. During the field trip, we implemented the projects in groups. After the field trip, the Aalto team and the partner team presented the results of the field trip at Universidad Modelo and are jointly completing the final project report. We will gain deeper experience in international collaboration while also strengthening their skills by exploring the Mayan culture which is integral to the identity and history of Mexico.

The community residents of EI20 who have direct contact with the team during the field trip: Around 30 local people living in EI20, primarily women, interacted directly with the team during the field trip. They participated in multiple interviews, participatory activities and workshops. Interviews with local residents also revealed mutual learning and knowledge sharing around student projects such as solar ovens, water purification, and water distilling. In the test of the solar oven, they collaborated with us and attempted to test traditional Mayan food using the solar oven. These collaborations provided the residents with knowledge and skills on how to use solar ovens, enhanced their confidence in independently operating solar ovens, and potentially reduced fuel consumption and smoke exposure, while also informing them about ongoing water purification projects in the community.

Final rights-holders:

The EI 20 Community: Approximately 420 residents are living in the community (Mexico PueblosAmerica, 2024). In an ideal situation, the ALM 2025 project will lead to a broader adoption of the solar oven, and improve user's health by reducing exposure to smoke compare to traditional cooking methods. In addition, ongoing water purification and solar water distilling projects aim to improve access to clean water, further supporting the community's health and resilience.

Relevant NGOs: From the first ALM project to now, the ALM project has continually acted as a catalyst in enhancing collaboration between the community's main related NGOs: Amigos de Calakmul, Design Your Action and Waterfinns. The process and results of this project may also help these organizations in their subsequent community work. At the same time, the collaboration system and experiences among the community, institutions and NGOs may also create potential chances for other NGOs to join and develop new work areas.

Other communities in Mayan cultural region: There are 80 other communities like EI 20 in the Calakmul area (ALM'24 Team, 2024). With the discovery of the ancient Mayan city Valeriana, this area expanded, and more communities are included in the Mayan culture area. If the adaptation of traditional Mayan cooking to solar ovens is successful in EI20, the community can share the knowledge with other communities. The other communities may also try to explore the possible integration of the solar oven in traditional cooking process, and may also benefit from health improvements. In addition, information about different methods for water purification, such as solar distilling, can be shared between other communities, which may further support these communities by improving their access to clean water.

One week before departure

In the week before our departure, we had a meeting with Kari Silfverberg, a solar oven specialist and the person who was part of the building process of the solar oven brought to the community by the 2024 team. This meeting was especially important because it was our first time seeing a real solar oven in person. Kari walked us through the details of solar cooking, answered all our remaining questions, and helped us repair and complete an almost-finished solar oven, just in case the one in the community was no longer functional. We briefly considered taking this newly finished oven with us to Mexico. However, after receiving reassurance from our Mexican partners that the existing oven was still in good condition, we decided to travel lighter and only bring a backup glass cover—the most fragile component. Since we couldn't find the right size in Helsinki, we ended up buying it later in Mérida.

During the last week before departure, there were also discussions to bring E. coli bacteria test kits from Aalto University to the field trip to test for presence of bacteria in the water samples of the solar distiller. However there weren't test kits available. Other possible materials needed for the distilling and interviews we trusted to find in Mexico, and therefore save space in our luggages.

Alongside our technical preparations, we made a packing list, started learning basic Spanish to support communication, and some of us even enrolled in a beginner's Spanish course at Aalto University. And then finally, the day arrived: we left the cold Finnish winter behind and boarded the plane, first to Frankfurt, then to Cancún. We were definitely a little nervous, but mostly excited and full of anticipation for the adventure ahead.

During the Field Trip

6 Arriving in Mexico and starting our projects in EI20

Upon our arrival in Cancún, we traveled by taxi to Tulum, where we spent two days adjusting to the local climate and time zone. This short break allowed us to recover from the long journey, ease into the new environment, and mentally prepare for the work ahead.

On March 17th, we continued our journey by bus to Mérida, where we spent the next two days at Universidad Modelo. It was exciting to finally meet the students from Modelo and UAM in person after weeks of online collaboration. While there was initial hesitation and some language barriers, we got-to-know each other quite fast. Planning the upcoming field trip together turned out to be more complex than expected.

On March 20th, we all took the bus together to the community of EI 20. After a long ride and arriving in the evening, we were welcomed with a delicious Mexican dinner, and then, tired but excited, we settled into our hammocks for the night.

In the following days, we connected even more with the other students and mentors. We shared meals, stories, and lots of laughs while learning about each other's cultures and backgrounds. A memorable moment was introducing the Mexican students to Finnish licorice: their reactions ranged from curiosity to playful horror. Group activities in the morning and reflection sessions in the afternoon helped us open up to one another even further. But we didn't only grow closer as a team, we also started building relationships with members of the local community. Through small interactions, gestures, the Spanish phrases we had learned, and involving them directly in our projects, we gradually developed a sense of trust and connection.

Besides the social and cultural exchanges, we put significant energy into our project work. As we gained new insights through our applied research methods, our project focus evolved, leading us to adjust our fieldwork plan and daily activities multiple times throughout the week. We've summarized all our team's activities, including the solar oven, solar distiller, and perception project, in an overview (Table 1). The next section will go into more detail about the applied methods used in our fieldwork.

Focusing now specifically on the solar oven project: our original goal was to adapt full traditional recipes for solar cooking. However, through hands-on testing and interviews with community members, we shifted our focus. We began identifying specific steps in the cooking process such as pre-processing ingredients or preparing simpler components, where solar ovens could offer the most benefit.

The project on solar distilling kept its original focus, to design, test the built prototype and take samples for further laboratory testing that couldn't be done in the field. Since we weren't able to get test kits to test for bacteria to take with us to the community, that part was then left out of the plan.

The plan to conduct the perception interviews took place in Mexico, right before going to the community, when the plans for the water distilling project were being finalized. The idea and then the plan for the perception interviews were put together based on discussions with the project mentors, to tie together the projects our Aalto team members were involved in and hopefully the insights from the interviews would be useful also for future projects.

Date	Place	Activities
18 March	Modelo University, Merida	Fieldtrip & Project preparation
19 March	Modelo University, Merida	Fieldtrip & Project preparation
20 March	Bus to El20	-
21 March	El20	4 Interviews, Solar cooking testing (Egg, Tortilla, Corn, Potato), Setting up distilling device, pH testing and sample taking
22 March	El20	2 Interviews, 2x Participation in traditional cooking, Solar cooking testing (Salsa), Setting up distilling device, pH testing and sample taking, Participation in construction workshop I
23 March	El20	Solar cooking testing (Eggs, Tortillas, Jamaica), Final workshop preparation, Setting up distilling device, pH testing and sample taking
24 March	El20	3 Interviews, Solar cooking testing (Salsa, Tortillas, Camote), Participation in construction workshop II, Final workshop preparation, Setting up distilling device, pH testing and sample taking
25 March	El20	2 Interviews, Final workshop
26 March	Bus from El20 to Merida	-
27 March	Modelo University, Merida	Presentation preparation
28 March	Modelo University, Merida	Final presentation

Table 1: Overview fieldwork activities regarding our projects in Mexico

7 Our project methodologies

During our fieldwork in El 20 de Noviembre, our team employed a combination of research methods, including interviews, participant observation, participatory activities, and experimental techniques such as solar oven testing and solar water distillation. This section outlines the specific methodologies used across our projects related to solar cooking, water purification, and community perception.

In collaboration with the community, we have adopted a variety of methods to explore the potential of integrating solar ovens into traditional Mayan cooking practices. We interviewed people in the community to learn about their experiences and ideas related to solar ovens as part of our methods. Moreover, we engaged in participant observation through the traditional cooking process to understand the culture and ways of cooking within the community. Through cooking tests using the solar oven, the cooking possibilities and time of different foods being cooked in the solar oven were recorded. All our experimentation was carried out using the solar oven constructed by the ALM2024's Aalto team and left in the community (Figure 4). Furthermore, some participatory activities, such as the workshop, were also used for data collection.



Figure 4: Solar oven testing in El 20 using the model built by Aalto students in 2024 and left in the community

The solar water distillation project involved the designing, construction, and field testing of the solar distilling prototype to evaluate its efficiency in the community's conditions. Data collection was done by testing the quality of the water in the field by pH testing, reporting the results and taking samples for further testing after the field trip.

Alongside with technical testing and solar cooking, qualitative perception interviews were conducted with community members to gather insights toward student projects, in this case the solar oven and water purification efforts. These interviews were carried out with the help of translators and aimed to gain perceptions of the usefulness of the projects and hear thoughts and recommendations from the community's perspective.

Together these methods in our team's projects provide understanding of the technical, cultural, and social aspects of implementing solar-based solutions in the community of El 20 de Noviembre. Below the methods used in our projects are explained further.

Method 1 - Interview about solar cooking and community members' perceptions

Interviews about solar cooking

Before we reached the El20 community for on-site interviews, we developed a preliminary interview outline based on our limited knowledge of the solar oven cooking and the traditional Mayan cooking process. Our core research questions were the interviewees' daily cooking habits, fuel usage, cognition and experience of the solar oven, problems encountered during use, and future interests. Through these research questions, we gained insights and demands regarding the use of the solar oven in the community, identified obstacles to their practical application, and explored opportunities to integrate the solar oven into traditional cooking practices.

Four interviewees were selected based on their connections with the solar oven in the community. The interviewee 1 (Figure 5) was one of the participants in the solar oven workshop in 2024. Her family has kept the solar oven that Aalto's team took to the community last year. The interviewee 2 (Figure 5) is a student in the community. She borrowed a solar oven from interviewee 1 several times and studied how to use it by herself.



Figure 5: Interviewee 1 (left) and interviewee 2 (right) were interviewed about the solar oven

The interviewee 3 (Figure 6) is a master baker in the community and he has rich experience in building ovens with limestone. The interviewee 4 (Figure 6) is almost one of the community members who know the ALM project best. She has heard of the solar oven but has not seen and used it yet. The interview was semi-structured, involving prepared questions that we flexibly adjusted based on each interviewee's unique circumstances and cooking experience.



Figure 6: The limestone-ash ovens (left) made by the interviewee 3 and the interviewee 4 (right) was interviewed in her hammock

When the interviewees already have some experience in using the solar oven, we focused on the actual challenges they faced when using the solar oven. For the interviewee 1, who has already had experience in the solar oven workshop, we interviewed her experience of cooking independently with a solar cooker and what benefits it has compared to traditional cooking methods. Together with the interviewee 2, we discussed the problems she encountered during her attempt to dry pet food using the solar oven, as well as her actual experience in using the solar oven. As for the interviewees 3 and 4 who have never seen or used a solar oven, the focus of the interview was on their interest in learning about solar ovens. We discussed with one of the interviewees her interest in using solar ovens and the health issues related to wood smoke.

Interviews about community members' perceptions of ALM projects

To gain understanding of the community members of El 20 de Noviembre's views on student projects in the community, interviews with residents of El 20 de Noviembre were conducted. The aim was to gather insights of community members' perceptions of previous and ongoing students' projects, particularly those related to solar ovens and water purification, such as solar distillation. The idea of the interviews was to especially focus on the community members' perceptions and gain understanding of, for example, how previous projects have been received, what has been useful, and what could be improved in future implementations. The method of the perception interviews is based on qualitative interviews. The interviews focused on two main topics, solar ovens as a cooking method introduced by previous student project (ALM 2024) and water-related projects, especially solar water distillation projects, such this year's and ALM 2023's project.



Figure 7: Petra (left) was interviewing and Valeria (right) was helping with the translation

Interviews began with questions about solar ovens and then transitioned to water purification. The interviews were conducted in local homes with the help of a translator, who translated the responses to English (Figure 7). In this project Valeria Canto Farjat, Julissa Traconis and Claudia Garduño García helped to translate the interviews, Valeria with four, Julissa with three and Claudia with one interview. In most cases, the interview included only the participant, who was interviewed, the interviewer, and the translator. Occasionally, other members of the student project were present to listen or contribute to ask some questions. Oral consent was received from all participants after explaining that participation was voluntary, and that the information would be used for a student project. One interview was joint with the rest of the Aalto team, since their project is related to solar ovens. The interview with the Aalto's solar oven team was recorded with the participant's permission. The remaining interviews were documented with handwritten notes with help of translation of the responses.

A total of 7 women and 1 man were interviewed. Participants were recommended by mentors, project staff, and community contacts who identified individuals with prior knowledge or involvement in student projects. The selection of participants was conducted with an aim to gain insights from members of the community who would have prior knowledge of the student projects or would at least to a certain degree, be aware of them or have been involved with ALM before. The only demographic data collected was gender, and the background of the participants was acknowledged, but not asked about specifically.

Method 2 - Participant Observation of Traditional Cooking Practices

The overarching research question guiding our participant observation was: How can solar ovens be integrated into traditional Mayan cooking practices?. To explore this, we engaged in participant observations of traditional cooking activities within the community. Our goal was twofold: first, to gain an embodied understanding of existing cooking processes and techniques, and second, to identify specific steps within these practices that might be compatible with or enhanced by solar oven use. While our original intention was to adapt entire recipes for solar ovens, these early observations helped us realize that focusing on specific components or ingredients might offer a more feasible and impactful pathway for integration.

We participated in two separate cooking sessions with community members (Figure 8). These sessions involved active participation, informal conversation, and visual documentation through photographs and field notes. The first session, focused on preparing panuchos (fried tortillas topped with chopped cabbage, pulled tomato, pickled red onion and avocado), was conducted without a translator, relying on gestures and basic Spanish. In contrast, the second session involved preparing a tomato soup and included the assistance of a translator, which allowed for deeper narrative engagement and the collection of personal stories related to the dish. This included questions such as: Do you often cook this dish? Is it connected to festivals or special occasions? Who taught you this recipe? Do you have any memories or stories associated with cooking it?. These stories provided insight into the embedded cultural value of the dishes.



Figure 8: Participation in traditional cooking sessions: making tomato soup (left) and preparing panuchos (right)

Method 3 - Solar oven cooking testing

Based on the dietary habits of the community, we selected ingredients that are staple in the community diet such as eggs, tortillas, corn, tomatoes and habaneros according to interviews and findings from participating in traditional cooking processes (Figure 9). We tested the cooking methods and durations of a total of 8 kinds of food in the solar oven to assess their feasibility. When setting up the solar oven, we exposed it to the maximum amount of sunlight and adjusted its orientation and reflectors' angle according to the position of the sun. For cooking containers, we chose an all-black metal pot that can absorb heat to the greatest extent. We ensure that the solar oven and the black pot maintain good functionality in every test to guarantee reliable results.



Figure 9: Testing process with eggs: placing eggs in a pot inside the solar oven before closing the lid (left) and the result of cooked eggs (right)

From March 21st to 25th, we conducted a series of cooking tests and recorded the process in detail (Table 2). Through these tests, we have collected rich quantitative and qualitative data. In each test, the exact time when the food was placed in the solar oven and taken out, as well as the state of the food at each stage, were all recorded. The data of these time points are crucial for analyzing the cooking efficiency and total cooking time of the solar oven. In addition, we also paid attention to the influence of other factors, such as the process of flipping the tortillas, the use of kitchen wares, and the amount of water used.

Method 4 - Participatory activities

During the testing process, we were fortunate to establish a good collaborative relationship with our host family members, and they brought us some ideas and inspirations for testing food. When we were conducting the solar oven cooking test, they showed great interest and took out the tortillas they made at home to try them themselves (Figure 10). After each test, we showed them the cooked food and collected their feedback to ensure that our results meet the tastes and standards of the community members. During the four-day test period, we also participated in two solar oven construction workshops, showcasing our test progress and results to community members and inviting them to the final workshop we held during the final day (March 25th).

Date	Food	Kitchen wares	Water	Time	Description
21.03.25	1 Egg	/	/	1 hour	Well cooked
	2 Eggs	in a black pot with lid	/	1 hour	Well cooked
	2 Tortillas (home made)	1 in a black pot with lid 1 out of the pot	/	2 hours	Tortillas were turned over after an hour, and well toasted after 2 hours
22.03.25	0.5 kg Corns	in a black pot with lid	500ml	5 hours	Corns were soaked overnight in pot, and after 5 hours cooking , some are easy to peel
	1 Potato	in a black pot with lid	500ml	5 hours	Potato were cooked with corns, and not cooked (65%) after 5 hours
	1 Potato	/	/	5 hours	Not cooked (75%)
	Salsa: 2 Tomatoes 1 Habanero	in a black pot with lid	/	2 hours	Well cooked
23.03.25	10 Eggs	in a black pot with lid	/	3 hours	Well cooked
	8 Tortillas (factory made)	/	/	2 hours	Tortillas were turned over after an hour, and well toasted after 2 hours
	5 Tortillas (home made)	/	/	2 hours	Tortillas were turned over after an hour, and well toasted after 2 hours
	Jamaica: 30g hibiscus flowers	in a black pot with lid	700ml	1.5 hours	Well done, added water and sugar to make the jamaica drink
24.03.25	Salsa: 4 Tomatoes 1 Habanero	in a black pot with lid	/	3.5 hours	Tomatoes and habanero were stripped; well cooked
	7 Tortillas (Home made)	/	/	2 hours	Tortillas were turned over after an hour, and well toasted after 2 hours
	7 slices of Camote	in a black pot with lid	/	2 hours	Not cooked
	1 slices of Camote	/	/	2 hours	Well cooked

Table 2: Solar oven cooking test results from March 21st to 25th



Figure 10: Host family member using the solar oven (left) and team member Ran showcasing test progress and results (right)

To further promote solar ovens, we invited community members to participate in the ALM2025 final workshop through multiple workshops, community broadcasts and community's WhatsApp group, allowing them to experience the actual operation and effects of solar ovens firsthand. Before the workshop began, we interviewed them three questions through sticky notes: their level of interest in using the solar oven, their understanding of the usage method, and their willingness to try to use the solar oven (Figure 11). Each question is measured with options ranging from 'not at all' to 'very much'. Subsequently, we introduced the assembly steps of the solar oven and encouraged them to assemble it by themselves. During the knowledge sharing session, we shared to the community members the cooking time, nutrition, health and other knowledge of solar ovens. After the sharing session, we invited them to taste Jamaica drink, salsa and tostadas cooked by the solar oven together, allowing them to experience the feasibility and deliciousness of solar oven cooking. After the workshop, we interviewed again with the same questions and compared it with the data before the workshop.



Figure 11: Community members assembling the solar oven (left) and answering the questions (right)

Method 5 - Solar water distilling

As part of the ALM 2025 project, our team also collaborated with UAM students on a project topic on water purification. The objective of the water purification project was to continue the work of the previous student projects, especially the water distilling project in ALM 2023. The aim was to test out the new design of a solar distilling device and its performance and yield in the community. The background of choosing the project topic was to provide ideas and solutions to the local challenges related to water quality and access to clean water, since the community is highly reliant on using rainwater. The community's water sources have high mineral content, which limits its use for drinking and other daily tasks, such as washing and irrigation. The main goal of the water distilling project was to provide the community a clean water source with decreased mineral content, so the distilled water could be used for irrigation purposes and washing.



Figure 12: Setting up the solar distilling prototype (left) and the final set up of the solar distiller (right)

The methodology of the solar distilling project involved the design, construction, and field testing of a solar distiller prototype that aimed to purify water using solar energy. Solar distilling mimics the natural water cycle with using evaporation and condensation as the key components in the process. When the sun rays heat up the water inside the distillation prototype, it evaporates and condensates back to liquid water on the plastic cover. In the evaporation and condensation process the impurities in the water are left to the bottom of the prototype and the condensation water is purified (Sampathkumar et al. 2010).

The students from UAM designed and constructed the prototype, and our team's (Aalto team) efforts were focused on the field testing. The prototype itself consisted of a slanted transparent plastic roof and wooden base, which are visible in Figure 12. The water to be distilled is poured into two aluminum trays located in the base, and when the heat from the sun rays heats up the inside of the device, the evaporated water will condensate on the slanted plastic roof, as seen in Figure 13, and run down to the sides to angled collecting channels that lead to a plastic bottle, where the distilled water is collected. The distilling device was checked before setting it up each day to add sealants, such as aluminum tape and silicon, to prevent leaks. The prototype was set to a sunny open space to maximize the amount of sun rays and heat exposure to increase the distillation yield.

The plan was to test different types of water found in the community, such as rainwater, well water, grey water from washing and water from the stream and take samples of the water before and after the distilling. The volume of the samples were measured and after the field trip they were sent to the laboratory in UAM for further testing of changes in the mineral content in the samples before and after the distilling. The samples taken to the laboratory testing were taken in sterile containers. In the field pH testing was done with CIVEQ pH test strips and the results were documented.



Figure 13: Condensation inside the solar distillation prototype

8 Project presentation to the partners

On 26th March, we took the bus back to Mérida from the community. Unfortunately, the air conditioner of the bus that came to pick us up broke down, so we had to take this mobile sauna all the way to Champotón. All along the way, we generously gave applause to every engineer who tried to fix the roof hatch to let the wind into the carriage. Over the following two days, all members of ALM25 gathered at Universidad Modelo to prepare our final presentation together. During this period, we encountered many difficulties. Thankfully, we have always been a team where members and mentors can communicate well with each other. Finally, we successfully delivered this presentation on Friday, March 28th (Figure 14).

On the weekend, we strolled in Mérida, watched the traditional Mayan fireball show in the square and explored Chichén Itzá. Compared with the modern and real Mayan life full of vitality in the community, the expression of Mayan culture in cities and scenic spots is to restore the ancient Mayan activities and relics exhibitions. No matter which way it is, it has enabled us to have a deeper understanding and recognition of the Mayan culture.



Figure 14: Presenting our project in Universidad Modelo

After the Field Trip

9 Analyzing our insights and results

We returned to Finland on the first day of April. The first week back was primarily dedicated to rest and recovering from the jet lag. During this time, we did some light work, such as digitising the information collected during the field trip and reflecting on our experiences. As we gradually recovered, we began to engage in more follow-up tasks such as analyzing the collected data. Our main insights are:

Solar Ovens in the community: high acceptance but insufficient knowledge

Through our interviews, we collected a wealth of qualitative data and found some commonalities among community members regarding their insights into the solar oven. All interviewees held a positive attitude towards the solar oven. One of the interviewees has successfully applied it to pet food drying and is satisfied with its effect. Even the interviewees, who have no experience in using the solar oven, have shown an openness and a willingness to try to use the solar oven. However, despite the positive attitude, the practical application of the solar oven is currently very limited. None of the interviewees has used it for cooking human food yet.

From the interviews, we realized that insufficient knowledge of using the solar oven has become the main obstacle preventing the local community from fully accepting this technology. Almost every interviewee mentioned that they did not know how to use the solar oven correctly, such as how to adjust the reflectors, how to judge the intensity of sunlight, or how to control the time etc. Furthermore, some people were worried that improper operation may lead to safety issues, such as burns, or that the food may not taste good. One of the interviewees tasted solar oven-cooked food in the previous year, but they thought the taste was not authentic enough.

In the interview, everyone expressed the need for more knowledge. Whether they want to learn how to use the solar oven or hope to understand how to make a solar oven by themselves, they all hope to have more demonstrations and practical opportunities in the future. Overall, the interviews made us realize that although solar ovens have a high potential acceptance rate in the community, to truly integrate them into traditional cooking practices, more educational support, cultural adaptation and technological optimisation are still needed.

Cooking traditional Mayan food by solar ovens is promising

Our analysis involved reflecting on the step-by-step processes observed during traditional cooking and assessing which elements could realistically be adapted for solar oven use. We focused on both technical feasibility and cultural appropriateness. Through our direct involvement in the cooking of panuchos and a traditional soup, we developed a clearer picture of where solar ovens could complement or enhance existing practices, and where they might fall short.

One of the most promising areas for solar oven integration emerged from the corn processing stage (Figure 15). In preparing panuchos, corn kernels were boiled with lime (cal) for several hours the day before the actual cooking, in order to soften them and remove the outer hulls. The method is energy-intensive and not time-sensitive, making it an ideal candidate for solar cooking. Because the corn is prepared in advance, the longer cooking time required by solar ovens does not conflict with immediate meal preparation needs.

Another step with high potential for solar adaptation was the preparation of salsa. We observed tomatoes and chillies being boiled before being blended into salsa (Figure 15). This is a simple process that requires moderate heat and minimal supervision. Moreover, salsa is a staple component across various meals, meaning that integrating the solar oven into this routine step could increase its everyday utility and relevance.



Figure 15: Dried corn used for tortillas (left) and tomatoes being cooked for salsa (right)

In contrast, some cooking tasks clearly presented challenges for solar oven use. The soup preparation was described by the cook as a “quick and easy” dish, often made in just 20 minutes quickly for her children or husband. The speed and immediacy of this meal are essential to its role in daily life, and therefore incompatible with the slow heat buildup of solar ovens.

Frying tortillas for panuchos (Figure 16) was another process that proved impossible to integrate into a solar oven. The traditional method involved heating oil to a high temperature and spooning it over the tortillas to create the desired texture and air pockets.



Figure 16: Frying tortillas for panuchos during the traditional cooking process

Solar ovens reduce smoke, save water, and speed up drying

The analysis results of the collected data in cooking testing reveal the effectiveness of the solar oven (Table 2). We observed that the oven performed exceptionally well in boiling eggs. Without adding water, whether it was one or ten eggs, they were all cooked within a reasonable time. The tortilla needs to be turned over once during cooking to prevent one side from getting burned. It only takes two hours to achieve a crispy effect. Salsa made with tomatoes and habanero, and Jamaica drinks were also very successful. These results prove the advantages of solar ovens in cooking food. They can avoid generating a large amount of smoke and save water at the same time. It can even shorten the time when drying tortillas. In addition, the flavor of the food cooked in the solar oven has also been tried by the community members, and we ensure their authenticity.

The workshop significantly enhanced the interest, knowledge and confidence of community members in using solar ovens

The sticky notes activity in the final workshop (Figure 17) showed that the participants have significantly improved in terms of interest, knowledge and confidence in use after the workshop (Appendix 1). Most participants showed a certain degree of interest in using solar ovens. However, before the workshop began, most participants still lacked knowledge on how to use the solar ovens, which was a major obstacle preventing them from using this technology more widely. After taking part in the workshop, some participants expressed that they were ready to use the solar oven. Participants recognized that solar ovens could reduce reliance on firewood, while reducing oil usage and preserving the nutrients of food. These benefits increased their interest and acceptance of this technology. In addition, the community members also showed interest in using solar ovens to cook foods such as fish, chicken and bread. The entire process, from the sticky notes interviews, hands-on practice, to result sharing, not only enhanced the community members' knowledge and understanding of solar ovens but also provided a strong basis for the further promotion of solar ovens in the community.



Figure 17: Final workshop sticky notes activity showing increase in knowledge and confidence in using the solar oven

Solar distilling results in numbers

Solar distillation tests were conducted between March 21st and 24th using different water sources. We had originally planned to test distilling on three different types of water, rainwater, well water and grey water, to find out how the distilling changed the mineral content in the different types of water. After all, we ended up testing rainwater from two different sources, well water and water from the stream in the village.

Distillation times ranged from 7 to 24 hours. The first distilling was the shortest, 7 hours, but since the yield from this distilling time was significantly small, our team decided to lengthen the time to 24 hours, so we left the device overnight and took the sample and pH-test on the following day of the set up.

For all water samples, pH levels decreased slightly after distillation. Most significant drop was the first rainwater sample, which initially had a high pH of ~10–11, that dropped to approximately 5.5 after distillation. Similarly, well and stream water and rainwater 2, all had pH values around 6–7, that also decreased to around 5 after distillation.

The total volume of distilled water varied. The highest volume recorded was distilled on March 23rd, when the total yield was 150 ml of distilled water collected after 24 hours of distilling. The smallest amount was collected on March 21st, when only about 15 ml (around the amount of one spoonful) of water was collected after 7 hours of distillation. The other distilling tests yielded about 50 ml of distilled water over a 24-hour period. The amount of input water to the device was around 2 liters for each distilling. The results from the distilling are collected in Appendix 2. The analysing of the results from the water samples' further laboratory testing is yet to be completed and therefore they are not included in the results at this stage.

Solar distilling has potential

The results of the solar water distillation tests show that the technique is capable of producing distilled water from different sources, even though the output volumes and efficiency varies. There are multiple factors that influence the efficiency and variations in the results.

One of the most noticeable variations between the tests was the output volume. The variations may be due to changing weather conditions during the field trip, especially the sunlight's intensity and temperature, which directly affect the amount of evaporation occurring inside the solar distillation device. One affecting factor is also the distilling time. The 7 hour long distilling test on March 21st, for example, had less distilling time and sun exposure, which could be reasons for the small output amount. Also after the first day of distilling (March 21st) we discovered leaks on the collecting channels and in the bottom of the device after taking it apart at night, which most likely have had an effect on the output amount.

After discovering the leak, we sealed the device better using aluminum tape and silicon to prevent it happening again. Also the bottom of the distilling device prototype is made of wood, which allowed the leaked water to leak into the wood and possibly all the way to the ground. If the material of the bottom would be waterproof, the leaked water from the collecting channels could be evaporated back up and collected. The reason there were leaks in the collecting channels was that the joints of the channel in the corners of the prototype had leaked, which we then tried to fix before the other distillations.

The drop in pH after distillation in every distilling test is an interesting result. The drop in pH indicates that there has been removal of alkaline substances from the water, and the result is reasonable, since distilled water is known to have low pH (5.6 ± 0.3) (Sun et al., 2022). Concern regarding the low pH of the distilled water is that is it safe for drinking since regular drinking water has pH around 6-7. The distilled water would be perfectly fine to be used in other daily tasks in everyday life in the community, such as bathing, washing clothes and dishes and watering plants.

For future improvements, several aspects could enhance the performance of the solar distillation system in the future, especially keeping in mind that the distilling would be done in EI 20. Improving the insulation and sealing would help minimize heat loss and increase the amount of evaporation and therefore the condensation and output. Optimizing the surface area and orientation of the distilling device could increase exposure to sunlight during the day. Although the amount of input water was around 2 liters in each distilling test, and the water area covered the bottom of the device effectively, increasing or decreasing the amount of water could have an effect on the yield.

Thoughts about the testing process itself are for example, conducting tests over a longer period of time and during different seasons of the year, although during rain season the need for water is not as acute but it would still be useful testing to collect data on how environmental conditions impact the performance. Overall, while the current prototype shows promising qualities and technique, it still needs some refinements and modifications for it to be fully functional and useful for the people in the community.

Insightful perception interviews: Solar ovens are seen as potential cooking method option

Among those who were familiar with the solar oven projects, either this year's or last year's (ALM 2024), the general perception was very positive. Community members saw solar cooking as a useful and sustainable alternative that reduces the need for firewood and indoor smoke, which are both important issues for households, since fuel for cooking can be expensive or difficult to access. There was also a point raised by one interviewee, that there is entrepreneurship in the community, and they could be interested in learning more about solar ovens and how to utilize them.

Several participants appreciated the opportunity to cook without having to check on the food constantly, since leaving the food to cook in the solar oven would free up time to do other things during the cooking process. The long cooking process would also add calculating when the meal would be ready, especially for the husbands when they come back from work for lunch and dinner.

When asked if the participants would like to build a solar oven themselves or have one ready made, there was interest in learning to build and maintain solar ovens themselves, especially if local and easily available materials could be used. The availability of the materials and easy access to them was highlighted by many participants, since many people in the community don't own cars or have the resources to go buy materials from further away.

When asked about how to motivate community members to use solar ovens, the importance of clear instructions and demonstrations were highlighted. They gave recommendations that the full cooking process with a solar oven should be shown rather than just presenting finished meals or hearing about the results afterwards. There was also an emphasis on encouraging spreading the information and learnings within families and the community, so more people would know about the solar ovens and their possibilities.

Recommendations regarding the usability of the ovens was to raise it from the ground for better operating, because the cooking process involves lifting objects in and out of the oven, and also securing the side reflectors and cover better, so wind wouldn't knock them down. The plastic cover also becomes hot, so adding some handles would help prevent burning hands when taking it on and off.

Insightful perception interviews: Community members wish for water-related projects to continue

On water-related projects, community awareness varied more widely. Only one participant recognized the previous solar distillation system specifically and knew more specifically about the method itself. Others had general knowledge of water purification efforts, such as the sand/soil filter project from earlier years (ALM 2018) and water quality testing that has been done before as part of previous projects (such ALM 2015, 2017 and 2023) but it was not clear which projects specifically they were thinking of. Despite the fact that the participants weren't aware of the water related projects specifically, there was strong support for the students to continue working on water-related topics. During the interviews, couple interviews also pointed out that the previous student projects have provided the community with valuable information about the quality of the water in the community.

When asked about the community member's interest in building solar distillation devices, the interest was relatively low, mainly due to concerns about the complexity of the device or the availability of the material. There was some interest to build the device, so then the community would have knowledge of how it works and skills to repair it if needed. Mainly the participants expressed interest in having one ready-made and thought how to use it with provided instructions. There was some concern relating to the materials of the distilling device, regardless of if it was self-made or ready-made, that if it would be heavy to move around. Reasons it would be needed to move around, would for example be, to move it inside or to shelter when it rains, or if it would need to be moved around to maximize sun exposure.

Several interviewees acknowledged the ongoing need for clean drinking water in the community and pointed out that issues like water "hardness", high mineral content, which is due to the presence of mineral-rich soil, is a remaining challenge in the community. When asked about recommendations or wishes for future projects, many participants mentioned the challenges that come with the hard water, since it can't be used for drinking, washing or irrigation purposes. The issue is not that they don't have water, but not clean water, so water purifying projects were wished to continue. Other recommendations for the future projects were cleaning the local stream of trash to reduce flood risks and returning the environment surrounding the stream more usable and enjoyable.

10 Communicating and reporting our projects

The next step included preparing materials for various communication tasks, reports and presentations for different audiences and stakeholders. We also remained in close contact with the ALM25 team through a few meetings to coordinate our ongoing work.

Meeting with one of the first members of ALM

A few days after the field trip, we had a meeting with Anni Hapuoja, who has been involved in ALM projects from the project's early stages. We introduced and presented our work to her. She provided us with valuable feedback and constructive suggestions, such as thinking about the impact over a longer period, like 30 years. She also facilitated a workshop with us to reflect on our roles, visions, and experiences throughout the process.

Mid review presentation

The mid review presentation took place two weeks after the field trip. We presented our progress to a diverse audience and received insightful comments and feedback. Attendees included SGT students and staff, Aalto University faculty, and other project collaborators.

Website content

As a part of the SGT studio projects, we contributed content to the official website, including a short descriptions and representative photographs. The content will be accessed at <https://sgt.aalto.fi/>.

Cookbook-Zine

Creating a cookbook was originally one of our project's former objectives, but the shift in the testing approaches made us change the objective from a full cookbook to a zine. The change of the test dishes from "big dishes" to "small dishes" also determined the change. The recipes and tips information collected from the dishes we tried was more appropriate for a zine and not enough for a full cookbook. Moreover, compared to cookbooks, the cost of zines are significantly low, and they can be easier to make multiple copies and distribute. The recipes and tips information collected from the dishes we tried was more appropriate for a zine and not enough for a full cookbook. Based on the test results, cooking information, and feedback collected, we created a zine titled "Solar Cooking Recipes and Tips" (Figure 18; Appendix 3 & 4). This zine includes instructions for using the solar oven, along with cooking recipes and tips we proved from the cooking test. The zine also includes a page dedicated to information of the benefits of solar cooking. The benefits include insights from the perception interviews with the community members. It serves as one of our final deliverables and is intended for practical use by the local community and later projects. It will be available in both English and Spanish versions to facilitate access to information for ALM members, community members and other stakeholders.

Final event

The SGT Final Event will take place on May 26th. Now, we are also preparing for the final event, which will mark the end of our project and showcase our team's efforts and outcomes. During the event, we will deliver a one-minute pitch. A student panel discussion will also be held. The event will be a public event and will be supported by marketing messages, posters and brochures.



Figure 18: The printed cookbook-zine *Solar Cooking Recipes & Tips*

10 Reflecting on our challenges and learnings

The most significant challenge we encountered was not technical, but communicative. The project involved multiple teams with diverse linguistic backgrounds and communication styles. Our Mexican partners mainly spoke Spanish, while we could only use English and several simple Spanish phrases. These language differences led to early misunderstandings and sometimes affected the team's mood. Big challenge was also the lack of communication services in the rural area, where there was no internet or cellular services to contact other people or coordinate the work there efficiently.

With support from the mentors and bilingual students, some of the communication challenges got some initial solutions during the university-based work. However, we realised that relying entirely on translation severely limited our interactions with local people. As we moved deeper into the community context, we made a deliberate effort to communicate directly with local people. We expanded our Spanish vocabulary and tried to use a broader range of communication strategies, including visual materials, gestures, and emotional tone, to bridge the language gap. We tried to communicate and conducted simple interviews and shared information in Spanish. After several successful presentations with photos, Spanish words, gestures, and humour. We received warm feedback and even some appreciation for our effort. These experiences pushed our team to become more adaptive, patient, and confident in handling different communication patterns.

The different lifestyle and culture also brought some small challenges. In Mexico the lifestyle is not as highly paced than what our team is used to, both in the city and the rural community. The course from Aalto part required us to work with a tight schedule and efficiency. During the early meetings, combining to get a general plan for both our team and Mexican partners was difficult, and we were frustrated by that. With more communication and working process, we gradually found the harmony of the pace for collaboration. During the time in the community, we lived together with Mexican students. In the jungle house, our nightly chats and jokes created a sense of belonging we didn't expect. We were exposed to more real local culture. Living closely with them brought unexpected warmth and laughter.

One of the primary challenges we faced was the integration of the solar oven into the existing cooking routines. In the families we worked with, cooking was mostly the responsibility of women, many of whom were already highly skilled in traditional cooking practices. Although from the interviews, people we talked with were interested in the solar oven, there was also considerable skepticism. Weather conditions and the need for continuous sunlight are its inherent shortcomings.

Moreover, the actual circumstances in the community and the actual test results were different from what we expected before we went to the community. As a result, we had to change some parts of the plan, even the objective, including discarding some of the workshop materials we had prepared.

However, these challenges turned into valuable learning experiences. We realized that co-design and ongoing communication with people were crucial for building trust and encouraging acceptance (Figure 19). We also learned not to overthink or be overwhelmed based on assumptions. What the local people think, need, and value are always the foundation of any kind of action and work.



Figure 19: Community members and project members communicated and tried the egg cooked from the solar oven (left) collaborating in preparing the corn to test in the solar oven (right)

A core principle we learnt throughout our work was the importance of collaboration and participation with the local community. We believe this approach proved essential not only in the solar oven project, but also in the solar water distillation work and the perception interviews. Engaging community members directly allowed us to better understand local needs, practices, and expectations, and fostered a deeper sense of trust and openness. We believe that such participation strengthens the long-term impact of the projects by encouraging local ownership, confidence, and knowledge around the introduced technologies. It's also a key factor in sustaining momentum for future efforts once our team departs. Our experience also showed us that communication is not only based on language, but also visions, emotions, feelings and much more.

11 Discussion and conclusion

Solar cooking

Before the field trip, we had high expectations for the solar oven. We hoped it could partially replace traditional wood-burning cooking and even planned to prepare some local signature “big dishes” with it, such as Pib (A traditional Mayan dish of marinated meat wrapped in banana leaves and slow-cooked underground) and Frijol con puerco (A Yucatecan stew of pork and black beans, typically served with rice and garnishes). However, these expectations began to shift during our initial observations and interviews.

Through discussions with the local people and internal reflections within our team, we gradually realized that the solar oven was not an appropriate or feasible replacement for traditional cooking. People seemed interested but unfamiliar with the technology, and were hesitant to give it a try. We found it puzzling, especially since a successful test with Pib had been conducted the previous year. We came to understand that the “big dishes” like Pib, though culturally significant, are not part of everyday cooking routines. Their special complexity may have attracted interest, but it discouraged practical use.



Figure 20: Tostada (tortillas) dried by solar oven

Based on these insights, we began to see the solar oven not as a replacement for traditional cooking, but as a supplementary tool. It is especially useful for cooking simple ingredients, such as eggs or drying tostadas (Figure 20). It is also a safe, smoke-free alternative in situations where fuel is scarce or where health concerns arise due to indoor air pollution from black smoke.

Our outputs justified this shift in our approach. The fieldwork experience helped increase awareness and interest in the solar oven as a more sustainable tool. It gave people more interest and motivation to experiment with daily cooking routines with a solar oven, also opening discussions about energy alternatives. Importantly, we also participated and witnessed the positive effects of intergenerational and intercultural knowledge exchange, which gave us a sense of accomplishment.

Overall, our analysis suggests that rather than attempting to adapt entire dishes, a more effective approach is to identify and target individual steps, particularly those that are time-flexible and culturally routine. By doing so, the solar oven can be introduced as a complementary tool for specific cooking tasks.



Figure 21: Community members sharing knowledge about traditional cooking (left) and our team member sharing the solar oven test results (right)

To evaluate our impact, we used a combination of qualitative and observational methods. We began collecting thoughts and feedback during initial interviews and continued documenting participants' thoughts and our interactions throughout the whole fieldwork (Figure 21). In the final workshop, we gathered reflections both before and after the fieldwork period. This allows us to conduct a comprehensive post-fieldwork analysis and better evaluate the impact.

Our impact is not simply measured by how often the solar oven is used. Instead of that, we focused on how our project influenced people's awareness and interest, whether they were willing to try the solar oven, and if they felt more confident using it. It is also worth considering whether our work and the whole ALM project fostered intergenerational and intercultural knowledge exchange.

Solar distilling

The solar water distillation tests carried out in El 20 de Noviembre provided practical insight into the performance of the prototype under real field conditions. While the design was functional, the results showed relatively low yields, since daily outputs ranged from as little as spoonful (around 15 ml) to a maximum of 150 ml over a 24-hour period. These results suggest that, although the solar distilling device is technically capable to work in the community's environment and context, its current design has limited efficiency and capacity to produce distilled water.

Variations in the performance of the prototype were likely influenced by several factors, including changes in weather conditions, sun intensity, and air temperature. In addition, the prototype's design itself may have impacted the results. Heat loss and leaking due to insufficient insulation or sealing could have reduced the amount of evaporation, while the angle and orientation of the device may not have been optimized to maximize solar exposure. The materials used in construction that were chosen for accessibility and simplicity can also affect the performance and efficiency of the solar distilling prototype.

The low water output and relatively long distillation times indicate that the current prototype would need some modifications before being a practical solution for household-level clean water needs. Improvements in design, such as increasing the sealing of the device, and potentially a greater surface area for evaporation and condensation, could increase performance. Ultimately, while technical outcomes from the solar distillation system were limited during our time in the field, the solar distilling project provided useful ideas and designs for future iterations of the method, and a clearer understanding of the limitations and potential of solar distillation in the community's environment.

Perception interviews

The perception interviews offered valuable context for how the community views the solar technologies introduced through student projects. Regarding the solar oven, the general response from participants familiar with the previous or current project was positive and the oven was seen as a sustainable option that reduces reliance on firewood and exposure to indoor smoke, which was seen as an important health benefit, since now there is a lot of smoke produced in the cooking processes that use firewood. Due to the smoke many have the kitchens separate and outside of the rest of the house. The decreased amount of smoke would make the living spaces more comfortable and healthy. Participants, of whom the majority were women, also appreciated the ability to leave food to cook unattended, allowing them to focus on other daily tasks, which could free up valuable time especially for the women in the community who do a lot of the families' house work.

At the same time, several practical concerns emerged. The need to time meals appropriately around the long cooking process was noted. There was a strong interest in learning how to build and maintain the ovens, but access to materials remained a key concern, as many community members do not have the means to travel outside the village to acquire them. Suggestions for integrating the ovens into the community were proving information and demonstrations of the use, but also improving the design for ease of use, such as adding handles to prevent burns.

In terms of water-related projects, interviewees generally lacked specific knowledge about earlier solar distillation efforts, though there was broader awareness of past water purification initiatives like sand filters and water testing. Altogether, participants expressed strong support for continuing projects to improve water quality.

The most frequently mentioned challenge was the "hardness" of water, which prevents the use of available water for drinking, washing, or irrigation. The stream in the village was also mentioned as a topic for future projects, with suggestions for cleanup efforts to reduce flooding and restore its environmental and social value.

Overall, the perception interviews reinforced the importance of designing projects that would be practically accessible for the community members. Community members' feedback highlighted both a willingness to engage in the student projects and the real-world constraints that affect adoption, such as costs of implementing new technologies in the community and the prejudices of the community members towards new technologies. Clear instructions, hands-on demonstrations, and knowledge-sharing were repeatedly identified as essential for successful implementation. These insights are crucial for shaping future efforts in EI 20 and ensuring that sustainable technologies are introduced in ways that align with community's context. It is important to note that the number of perception interviews was limited, meaning the insights provide valuable but not fully comprehensive reflections of the wider community's views.

12 Next Steps and imaging the future

During the field trip, we noted several ideas and opportunities for improvement that we hope future project teams can explore.

Improving the solar oven and increasing number

One major direction is improving the design and usability of the solar oven based on our test results and community feedback. During the field trip, our Mexican partner teams conducted several co-design workshops to share the information and build the solar oven together with the local people, in which the Aalto team also participated. Moreover, the Aalto team shared the test results and got some feedback from the local people. Although the solar oven has existed for a long time, its introduction to this particular community only began last year. Through daily use, we identified several design issues. For instance, the angle range of the reflectors and the cover is too limited (Figure 22, left), and the cover support is not stable enough and sometimes it falls due to slight vibrations and wind. Additionally, the current transparent sheet design makes it hard for a single person to use the solar oven; A better design could include a one-sided hinge with single-hand operation (Figure 22, right). As more community members engage in testing, more suggestions for improvement are likely to emerge. Timely updates to the oven's design will be essential to maintaining community interest and ensuring the oven fits local needs.



Figure 22: A set-up solar oven with reflectors, cover and transparent sheet (left) and team member Johanna simulating one-sided handle possibility (right)

We also hope more families will be engaged in solar oven testing, and more solar ovens will be brought or made in the community. After this year's final workshop, four families had received solar ovens, and interest remained high. Even after the ovens were distributed, some people stayed behind to ask questions and seek more information. This shows a clear demand. Increasing access to solar ovens is crucial so more people can try them out and learn through hands-on experience.

Further developing of zine or cookbook

The zine “Solar Cooking - Recipes and Tips” became the outcome of this year's project. If possible, we sincerely hope future teams can build on the zine we started, adding more recipes, ingredients with clear instructions for cooking times and techniques specific to solar ovens. Ideally, a more well-developed zine or cookbook could also include stories and cultural histories behind local dishes, preserving intangible heritage alongside promoting sustainable practices.

Broadening the use of community-based approach

Looking forward, we envision using the community-based approach developed in this project to identify other challenges that could be addressed with sustainable solutions. For example, issues related to water purification and waste burning were repeatedly mentioned. We observed that open waste burning is still a common practice (Figure 23). We even saw the waste produced during our activities burned just 20 meters from the host family's house. These observations could guide future topics that continue to foster community autonomy and ecological resilience.



Figure 23: Smoke from waste burning (left) and Waste burning site (right)

Enhancing project visibility in the community

One key observation from our field trip is the limited visibility of projects among community members, since almost only the families that have been involved or cooperated with the projects have a relatively high awareness of the projects happening in the community. Increasing the visibility of projects through local events would strengthen community connection and engagement, improving the relevance between the projects and local people.

Fostering collaboration and participation

Projects with more collaborative and participatory approaches can maximize the long-term impact. The later projects could try to involve more local residents, particularly women and youth. Increasing their power in decision-making and implementation of the future projects could foster ownership of the projects in the community in the long run. Most importantly, more collaboration and participation could ensure the projects and solutions are better aligned with the actual community needs.

Community visions for future projects

For the next steps in terms of topics for future work in the community, based on the perception interviews with community members, future project topics could focus on cleaning and treating existing water sources, particularly addressing the issue of hard water. There is also strong interest in cleaning the environment around the local stream, both to prevent flooding and to restore its use as a natural and social space. Projects that incorporate solar distillation or other water purification methods remain relevant, especially if improved for efficiency and adapted to local needs. Considering environmental wellbeing and using locally available materials in the projects are valued, as well as promoting open knowledge sharing, so that tools and practices can be maintained and taught within the community.

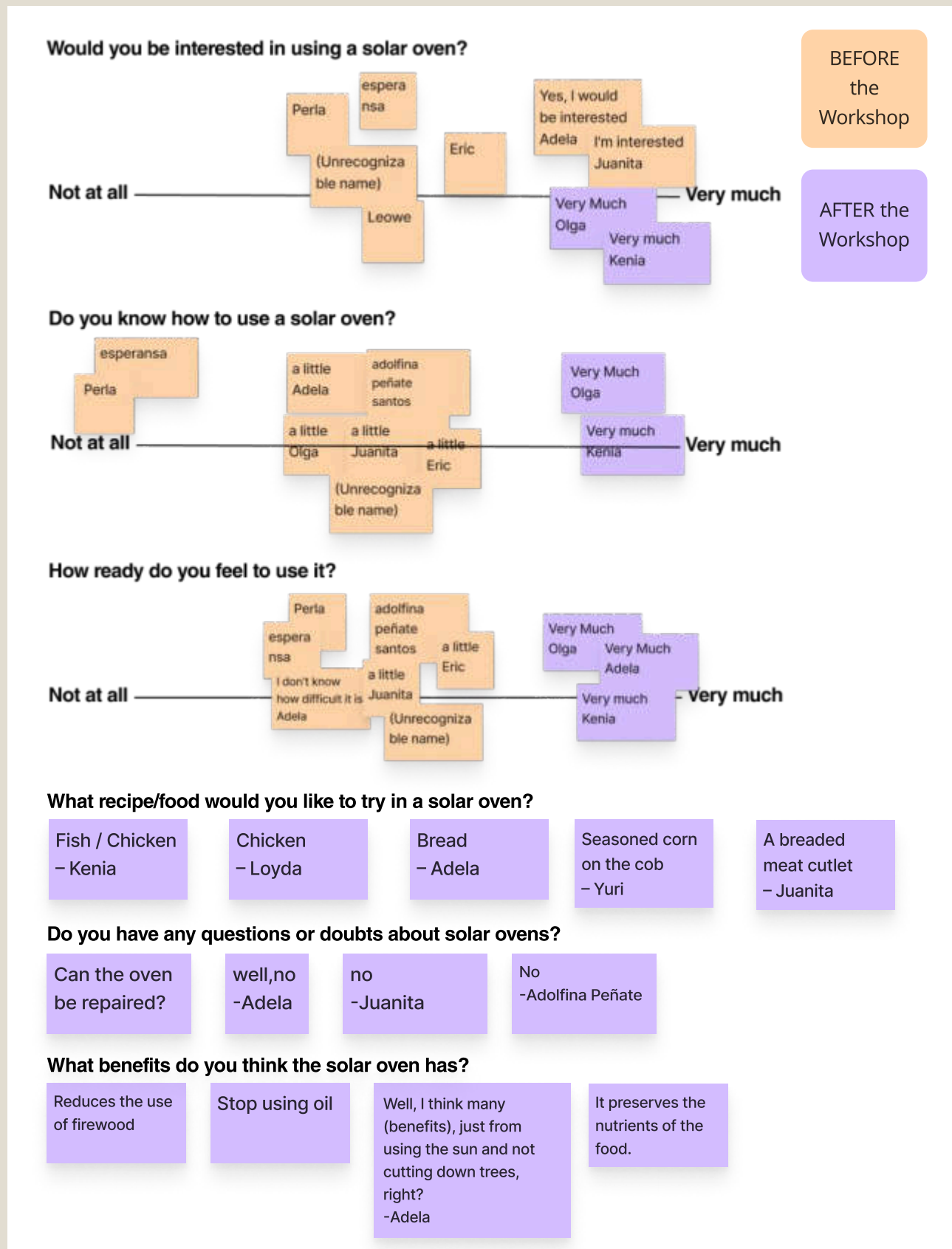
That was our “Muy bien” journey.



Adiós!

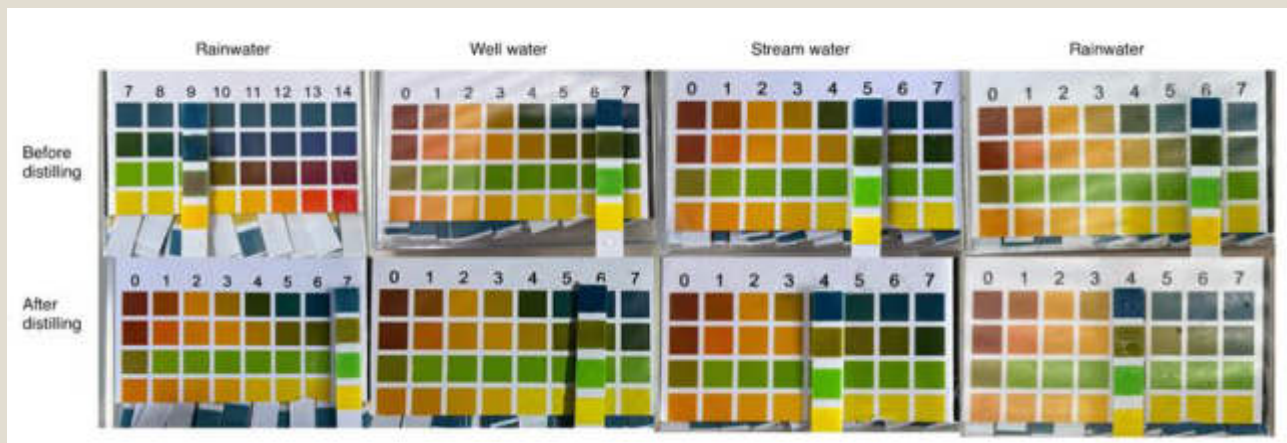
Appendix

Appendix 1- Final workshop sticky notes activity (own illustration)



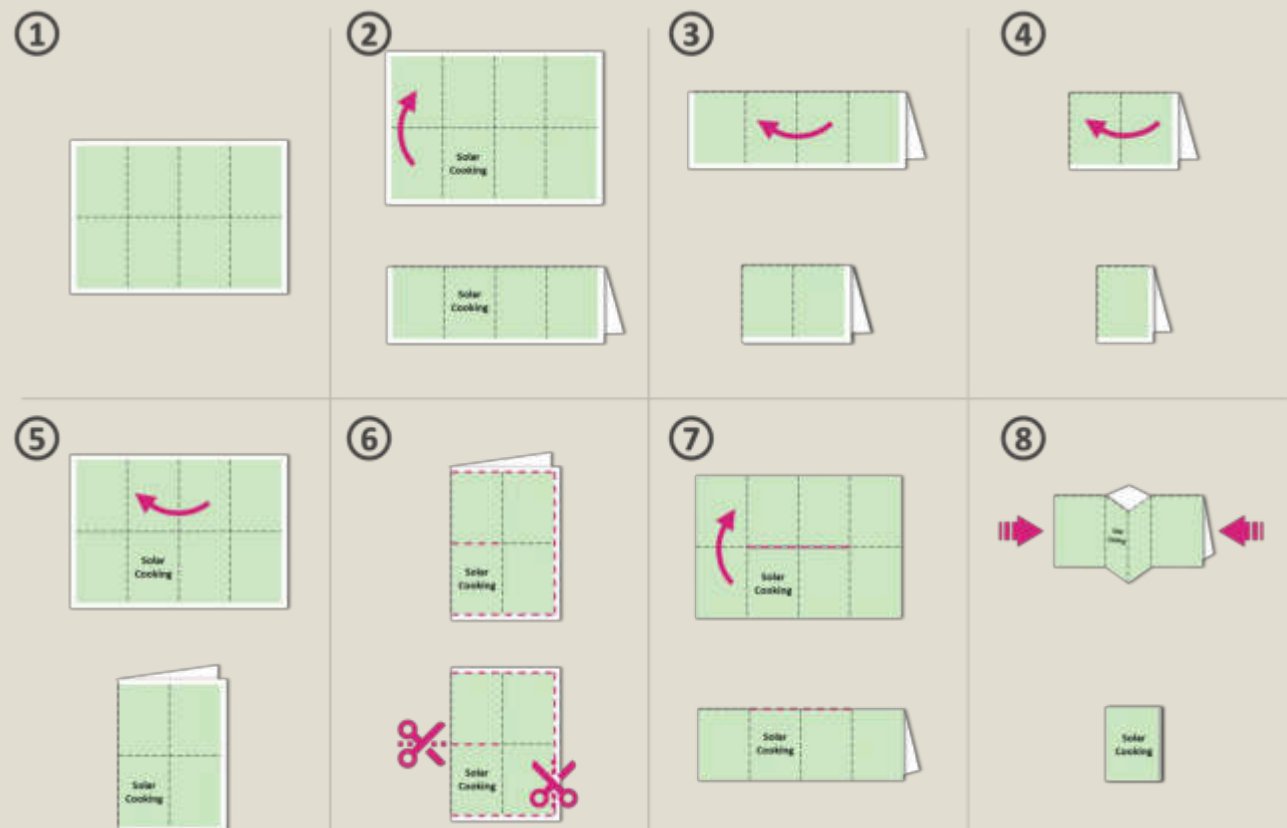
Appendix 2 - Water results (own illustration)

Date	21.3.	22.3.	23.3.	24.3.
Type of water	Rainwater	Well water	Stream water	Rainwater 2
pH before distilling	(10-11) ~10	(6-7) ~7	(6-7) ~6,5	~7
pH after distilling	(5-6) ~5,5	(5-6) ~5,5	(5-6) ~5,2	(5-6) ~5,5
Distilling time (h)	7	24	24	24
Sample taken before distilling (ml)	100	100	100	100
Sample taken after distilling (ml)	-	50	100	50
Total volume distilled (ml)	~15 (size of a spoon)	50	150	50
Temperature (°C)	32	32	33	33



Appendix 3 - Diagram Instructions for how to make a zine from one sheet of paper (own illustration)

It should be printed on A4-sized paper. After cutting the white edges, the folding method is as follows:



Appendix 4 - Our cookbook-zine in English and Spanish (own illustration)

Benefits

- No wood or other fuels are needed
 - Saves money and resources
- Reduction of smoke from cooking process
 - Increased air quality at home
 - Health benefits from reduction of smoke inhalation
- Spare time to do other things while the food is cooking in the oven
- Environmentally friendly
 - Uses only solar energy
 - No additional fuels and no smoke to the environment

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Jamaica

- Step 1:** Put the jamaica flowers in the black pot with right amount of water.
- Step 2:** Cover it tightly with a lid and cook in solar oven for: **2 hours**.
- Step 3:** Take out the pot, strain the liquid, add water and sugar to taste.



Leftover flowers can be used as a natural dye for coloring fabrics.

Salsa

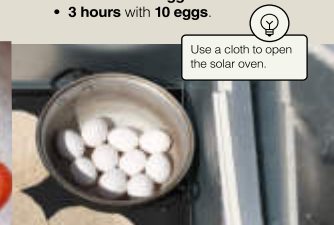
- Step 1:** Open up the tomato, poke holes in the habanero.
- Step 2:** Put them in the black pot without water, cover it tightly with a lid.
- Step 3:** Cook in solar oven for: **3 hours** for **3 tomatoes** and **1 habanero**.



Pair with solar oven tostadas and eggs to make chilaquiles.

Eggs

- Step 1:** Put the eggs in the black pot without water.
- Step 2:** Cover the black pot tightly with a lid.
- Step 3:** Cook in solar oven for:
- **1 hour** with **1 egg** or
 - **3 hours** with **10 eggs**.

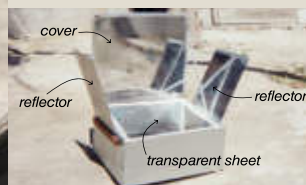


Use a cloth to open the solar oven.



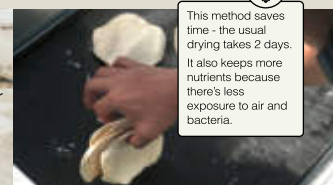
Set-Up

- Step 1:** Place the solar oven toward the sun, open the cover, and attach the two side reflectors.
- Step 2:** Open the transparent sheet, put in food directly or inside a black pot and close the sheet.
- Step 3:** Adjust the ovens to focus sunlight into the oven during all cooking time.



Tostadas

- Step 1:** Put the Tortillas directly in the oven, but ensure separation.
- Step 2:** Cook in solar oven for:
- **2 hours** to be completely dry.
- Don't cook other dishes containing water at the same time - the tortillas will not get crispy!



This method saves time - the usual drying takes 2 days. It also keeps more nutrients because there's less exposure to air and bacteria.

Beneficios

- No se necesita leña ni otros combustibles
 - Ahorra dinero y recursos
- Reducción del humo del proceso de cocción
 - Mejor calidad del aire en casa
 - Beneficios para la salud derivados de la reducción de la inhalación de humo
- Tiempo libre para hacer otras cosas mientras la comida se cuece en el horno.
- Ecológico
 - Utiliza sólo energía solar
 - Sin combustibles adicionales y sin humos para el medio ambiente.

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Jamaica

- Paso 1:** Coloca las flores de jamaica en la olla negra con la cantidad adecuada de agua.
- Paso 2:** Cubre bien con una tapa y cocina en el horno durante: **2 horas**.
- Paso 3:** Saca la olla, cuela el líquido, agrega agua y azúcar al gusto.



Las flores sobrantes pueden utilizarse como tinte natural para teñir telas.

Salsa

- Paso 1:** Abre el tomate y haz agujeros en el habanero.
- Paso 2:** Colócalas en la olla negra sin agua y cúbreala bien con una tapa.
- Paso 3:** Cocina en el horno durante: **3 horas** con **3 tomates** y **1 habanero**.



Acompaña con tostadas de horno solar y huevos para hacer chilaquiles.

Huevos

- Paso 1:** Pon los huevos en la olla negra sin agua.
- Paso 2:** Cubre bien la olla negra con una tapa.
- Paso 3:** Cocina en el horno durante:
- **1 hora** con **1 huevo** o
 - **3 horas** con **10 huevos**.

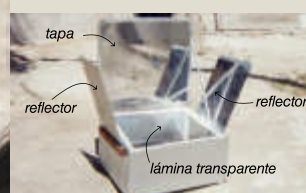


Utiliza un paño para abrir el horno solar.



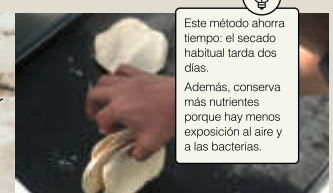
Configuración

- Paso 1:** Coloca el horno solar orientado hacia el sol, abre la tapa de horno y fija los dos reflectores laterales.
- Paso 2:** Abre la lámina transparente, coloca el alimento directamente o dentro de una olla negra y cierra la lámina.
- Paso 3:** Ajusta el horno para enfocar la luz solar hacia el interior del mismo durante el tiempo de cocción.



Tostadas

- Paso 1:** Coloca las tortillas directamente en el horno, pero asegúrate de separarlas.
- Paso 2:** Cocina en el horno durante:
- **2 horas** para que estén completamente secas.
- No cocina otros platos que contengan agua al mismo tiempo, ¡las tortillas no quedarán crujientes!



Este método ahorra tiempo: el secado habitual tarda dos días. Además, conserva más nutrientes porque hay menos exposición al aire y a las bacterias.

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Technology Statement

In terms of writing, the authors used assistance with the language for this paper. Both DeepL and Grammarly were applied to improve the author's original content, for paraphrasing, spell-checking, and grammar.